Statcast Swing Contact Analysis
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0. Abstract

This project analyzes how swing mechanics vary across pitch contexts and how these variations relate to offensive performance in baseball. Using Statcast pitch-level data, we examine how swing path tilt and attack angle differ across locations within the strike zone, and how swing path tilt is associated with contact quality metrics such as sweet spot rate, hard-hit rate, and expected wOBA. Exploratory visualizations and regression models indicate that swing mechanics adjust across pitch locations and exhibit nonlinear associations with contact quality, with performance peaking at intermediate swing configurations. We further outline a team-level analysis framework using LASSO regression to study how swing-related and batted-ball features scale up to explain run production. Overall, the project demonstrates how data preprocessing (tidyverse), visualization (ggplot), and modeling techniques (tidymodels) covered in this course can be applied to real-world sports data.

1. Introduction

1.1 Background and Motivation

Recent advances in player-tracking technology have made detailed pitch-level and batted-ball data widely available in professional baseball. Statcast data record pitch location, swing geometry, and contact outcomes with high precision, enabling quantitative analysis of how hitters adjust their swing mechanics across different pitch contexts and how these adjustments relate to performance. Since batting involves responding to pitches that vary in height, location, and speed, understanding how swing mechanics adapt to pitch context and whether such adaptations translate into better contact outcomes remains a central question in baseball analytics.

1.2 Research Questions

This project addresses four related questions discussed in sections 3-6:

- **Q1:** How does pitch location within the strike zone affect a batter's swing angle?
- **Q2:** How does pitch location influence a batter's attack angle, and is adjusting it by pitch type and zone important for good hitting?
- **Q3:** How does swing angle relate to contact quality metrics such as barrel rate, hard-hit rate, and expected wOBA?
- **Q4:** Can LASSO regression be used to identify which game-level features best explain a team's runs scored over a season?

2. Data description and Preprocessing

2.1 Data Source and Key Variables

The dataset used in this project is drawn from Major League Baseball's Statcast system [1], consisting of pitch-by-pitch observations. Pitch context is captured using horizontal and vertical plate coordinates. Vertical pitch location is normalized using batter-specific strike zone boundaries to account for differences in player height. Swing mechanics are measured using swing path tilt and attack angle, which describe the geometry of the bat path at contact. Contact quality is assessed using sweet spot events, hard-hit events, and expected weighted on-base average (xwOBA). For the team-level analysis, total runs scored is used as the outcome variable.

2.2 Data Cleaning and Feature Engineering

Data preprocessing was performed using *tidyverse* tools. Observations with missing values in key variables were removed. Vertical pitch location was normalized relative to each batter's strike zone, and pitches were categorized into meaningful regions within the strike zone based on both vertical and horizontal location. These zone definitions were used to support the exploratory analyses of swing mechanics across pitch contexts.

Additional derived indicators, including binary measures for hard-hit and sweet spot contact, were constructed to facilitate analysis of contact quality. For analyses involving swing mechanics and contact outcomes, data were aggregated at appropriate levels to compute summary statistics and model inputs. These steps ensured consistency across analyses and enabled clear interpretation of swing mechanics across pitch contexts.

3. Analysis: Pitch Location and Swing Mechanics (Q1)

3.1 Pitch Zone and Swing Path Tilt

For this section, we are taking a look at the effects of swing tilt on the outcomes of hitters, while also adjusting for the place the ball is pitched. This section is mainly here to check whether the skill of changing your swing path is important for the success of hitters, but I also plan on checking other factors to see if I can find any other noticeable trends in the data.

To do this, I took the data from players with at least 100 plate appearances over the 2025 season and using Statcast's bat tracking data, found their average swing tilt for 5 different zones in the strike zone.Top shadow includes the very top of the strike zone and slightly outside of it, and inverted for the bottom shadow, being the very bottom of the zone and just outside of it. Then, I also include the middle of the zone, the middle high part of the zone, and the middle bottom part of the zone. Getting all of these different zones gives us a good array of different swings each of the players took on average, and from this, I started looking for some trends.

Firstly, I looked at general performance. Taking the top 25 best hitters, according to wOBA, a general all purpose hitting statistic, I plotted this average in accordance to the league average swing to figure out whether there were certain types of swings that generally did better in certain zones.

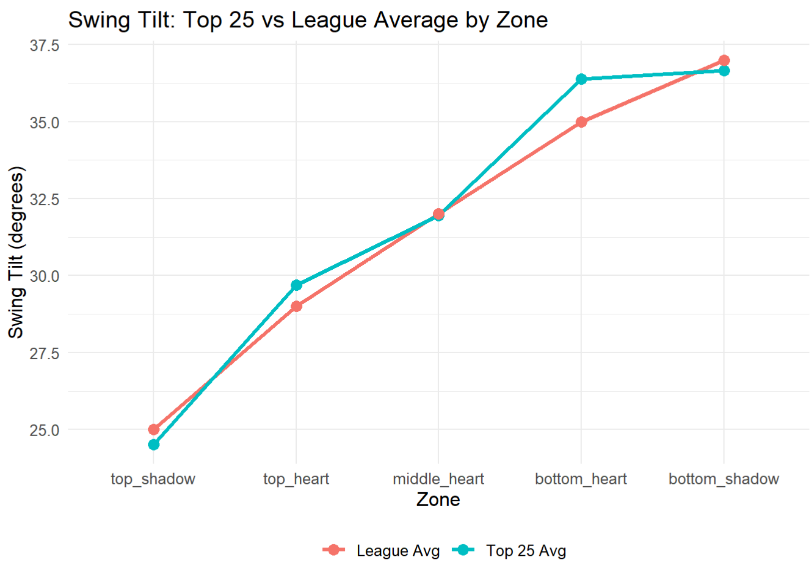

Figure 3.1: Top 25 Best Hitters in Each Zone vs League Average

One thing that I can see from this data is that the line very closely follows the league average hitter, only really deviating at the bottom heart of the zone with higher swing tilts being important for success in this category. This may lead us to believe that it may be important in this part of the zone, but nothing conclusive, so I did a general R test with the swing path tilt and each of the different parts of the strike zone. A positive result in this case would suggest more tilt being better in this zone, whereas a negative result would suggest better results with less tilt.

```
##   zone          metric               r
##   <chr>         <chr>            <dbl>
## 1 top_shadow    swing_path_tilt -0.0467
## 2 top_heart     swing_path_tilt  0.145
## 3 middle_heart  swing_path_tilt  0.0876
## 4 bottom_heart  swing_path_tilt  0.0986
## 5 bottom_shadow swing_path_tilt  0.0694
```

Table 3.2: R Value of Swing Tilt with wOBA

This general result is supported by the chart, where in 4 out of 5 categories a higher swing tilt seems to be useful for better results. This makes sense looking at our previous chart due to most of the line being over our average, but in all honesty, none of the factors presented show a strong correlation with results. We can also just look at some notable hitters, and see that there are successes with different approaches.

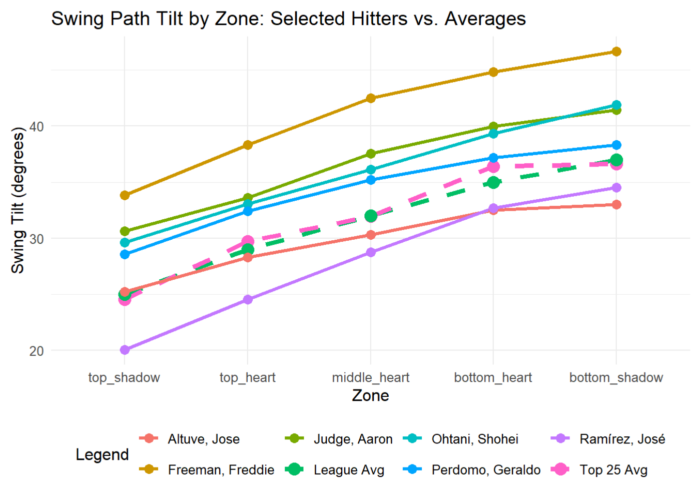

Figure 3.3: Swing Path Tilt of Notable Top Hitters

One of the only things we can really say about this data is that many of the taller players, Aaron Judge, Shohei Ohtani, have higher tilts, where notable shorter players like Jose Altuve and Jose Ramirez end up with lower swing tilts.

Now we have looked at general averages, that really only covers whether certain swing tilts are better for different parts of the zone. This doesn't cover the actual variance of the swing path tilt. So I sorted each of the players to have much they change their swing in accordance to where the ball is pitched. The highest swing tile variance players are on the left, while the lowest variance players are on the right:

```
## # A tibble: 15 × 3
##    player_name        avg_tilt_diff max_tilt_diff
##    <chr>                      <dbl>         <dbl>
##  1 Arraez, Luis                7.07         11.0
##  2 Farmer, Kyle                6.32          9.45
##  3 Bellinger, Cody             6.26         10.5
##  4 Marte, Starling             6.12         10.2
##  5 Call, Alex                  6.06          9.14
##  6 Verdugo, Alex               6.06          9.55
##  7 Hoerner, Nico               6.06          9.42
##  8 Alvarez Jr., Nacho          6.05          9.11
##  9 Ozuna, Marcell              6.04          8.73
## 10 Myers, Dane                 6.01          9.17
## 11 O'Hearn, Ryan               6.01          9.09
## 12 Sheets, Gavin               5.99          9.43
## 13 Chapman, Matt               5.98         10.2
## 14 Moore, Dylan                5.90          8.26
## 15 Young, Cole                 5.89          8.91
```

```
## # A tibble: 15 × 3
##    player_name        avg_tilt_diff max_tilt_diff
##    <chr>                      <dbl>         <dbl>
##  1 Reynolds, Bryan             2.37          4.50
##  2 Soto, Juan                  2.49          3.63
##  3 Schwarber, Kyle             2.79          4.28
##  4 Harper, Bryce               2.81          4.53
##  5 Jansen, Danny               2.83          4.20
##  6 Hamilton, David             2.86          6.35
##  7 Martinez, Angel             2.87          5.83
##  8 Morel, Christopher          2.93          3.87
##  9 Hassell III, Robert         2.95          4.88
## 10 Suárez, Eugenio             2.95          5.14
## 11 Wade Jr., LaMonte           2.96          5.08
## 12 Altuve, Jose                3.00          5.10
## 13 Fry, David                  3.04          4.62
## 14 Trevino, Jose               3.06          3.86
## 15 Marte, Noelvi               3.06          6.12
```

Figure 3.4: 15 Most Variance According to Zone

Figure 3.5: 15 Least Variance According to Zone

3.2 Interpretation of Results

What this data tells me is that the amount of tilt difference is important to the kind of contact you are looking for as a hitter. For example, players like Luis Arraez, Nico Hoerner, Starling Marte, Dylan Moore, and Dane Myers are all some of the higher batting average players in the game, but often trade their contact ability for power. For players in the low variance section, we have NL HR runner up Kyle Schwarber, 40 HR leader Juan Soto, alongside notable power hitters Bryce Harper, Christopher Morel, Eugenio Suarez, and Noelvi Marte.

From this, given my previous knowledge of these players, I have concluded that a lower variance allows players to swing harder, and therefore sacrifice contact quantity for contact quality, whereas high swing variance allows for a higher level of contact, trading contact quality for contact quantity. Keep in mind, however, that this skill is still important, as every major leaguer at least has some version of this swing tilt difference skill in their at bats, it is just how much the player does it that controls for this quality vs. quantity trait.

4. Analysis: Pitch Location and Attack Mechanics (Q2)

4.1 Pitch Zone and Attack Angle

To examine how pitch location influenced attack mechanics, we analyze the relationship between the pitch-zone location and batter's attack angle at contact. As stated before, the data was obtained and pre-processed to a .csv file. To account for differences in batter height and strike zone definition, vertical pitch location was normalized using each batter's recorded top and bottom of the strike zone.

Pitches were categorized into low, mid and high vertical zones based on normalized pitch height, with horizontal location classified as inside, middle or outside relate to the center of the plate. Attack angle is treated as the primary response variable and is summarized across pitch zones using distributional plots.

4.2 Interpretation of Results

Starting with the overall distribution of attack angles - we see a normal distribution with spread centered slightly above 0 degrees. This indicates that most batters tend to swing with a modest upward path rather than a steep downward chop. This aligns with modern hitting approaches that emphasize lifting the ball to generate extra-base hits. The long tail to the left suggests that extreme downcut swings occur less frequently but are still present due to power-hits.

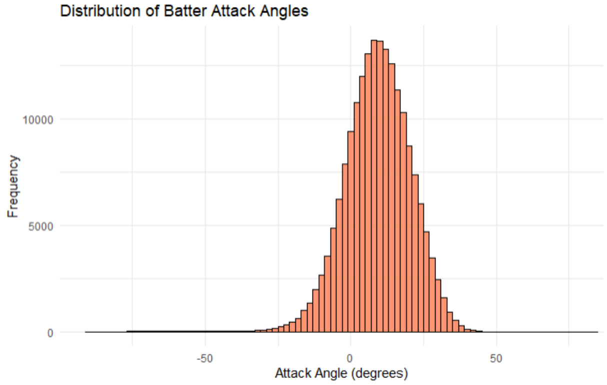

Figure 4.1. Distribution of batter attack angles.

z_zone	mean_attack_angle	sd_attack	n
Low	13.22	11.53	77161
Mid	7.41	11.21	58846
High	5.17	10.59	52699

Table 4.2. Summary statistics of attack_angle grouped by pitch zone.

After aggregating attack angle by vertical pitch zone, a clear pattern emerges as shown in the table below. Low pitches produce the steepest swings, with an average angle of 13.22°, compared to 7.41° for mid-zone pitches and 5.17° for high ones. The clear decline indicates that batters intentionally increase their swing plane on low pitches to lift the ball, consistent with our observations in baseball.

Despite differences in the mean, the standard deviation roughly hovers around 10-11° suggesting that while the average swing plane shifts with pitch height, individual swing variability is similar across regions.

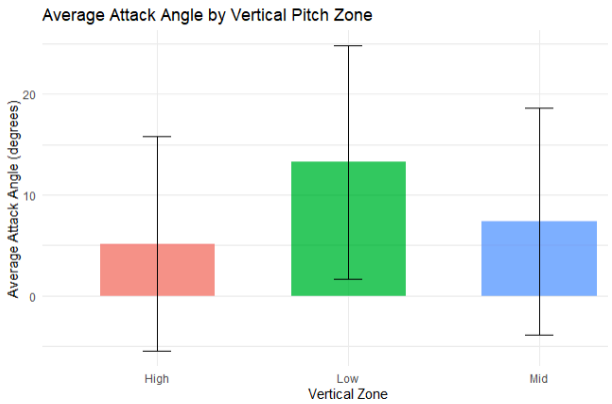

Figure 4.3. Boxplot showing average angle attack separated pitch zones.

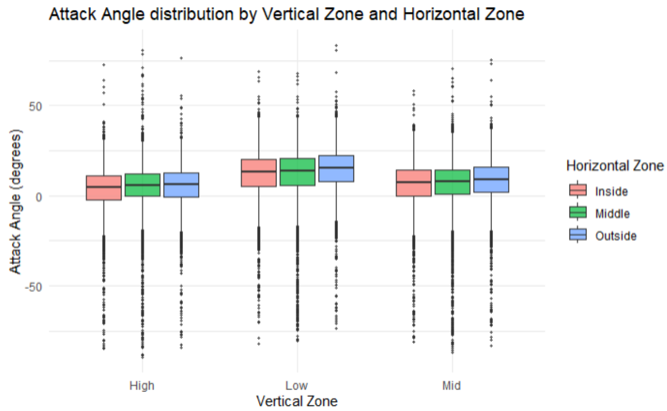

Figure 4.4. Attack Angle (raw) distribution by by vertical vertical and horizontal zones.

The boxplots further reveal how attack angle varies jointly across vertical and horizontal regimes. Hitters treat the inside pitches in the low zone with the widest spread, ranging roughly from 5° to 20°, suggesting more variability. Outside pitches cluster together around 10°-12°. In the High zone, attack angles compress across all horizontal zones, general staying between 3° to 17° showing less differentiation.

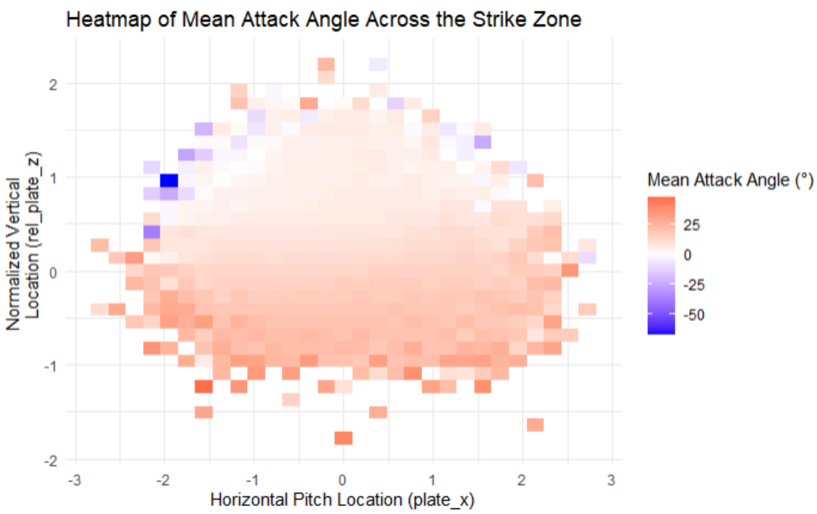

Figure 4.5: Heatmap showing attack angle by averaging each strike zone (normalized) bin.

Lastly, a heatmap shows how attack angle varies across the strike zone. The strike zone position was achieved by normalizing the x,y coordinates to account for height variations. In the upper part of the strike zone, attack angles flatten out, often dropping into the -20° to -40° range, especially on pitches high and outside. This reflects hitters chopping down or leveling their swing to simply make contact. In the middle of the zone, attack angles are steadier, sitting around 5° to 10°. Inside pitches show slightly higher values for a natural upward path for driving the ball. In the lower zone, hitters generate the steepest angles, frequently above 15° to lift low pitches and create loft.

5. Analysis: Swing Mechanics and Contact Quality (Q3)

5.1 Model Setup

To examine how swing mechanics relate to contact quality, we analyze three outcome measures: sweet spot probability, hard-hit probability, and expected weighted on-base average (xwOBA). Swing path tilt is treated as the primary explanatory variable, and quadratic terms are included to capture nonlinear relationships suggested by exploratory analysis. For sweet spot and hard-hit outcomes, logistic regression models are used, while linear regression is applied to xwOBA.

To assess whether pitch context modifies these relationships, interaction terms between swing path tilt and pitch height categories (low, mid, and high) are included. Model predictions are visualized as fitted response curves by pitch height to facilitate interpretation of how the relationship between swing path tilt and contact quality varies across pitch contexts.

5.2 Interpretation of Results

Across the three outcome measures, the relationship between swing path tilt and contact quality varies substantially by pitch height, indicating that mechanically favorable swing configurations depend on pitch context.

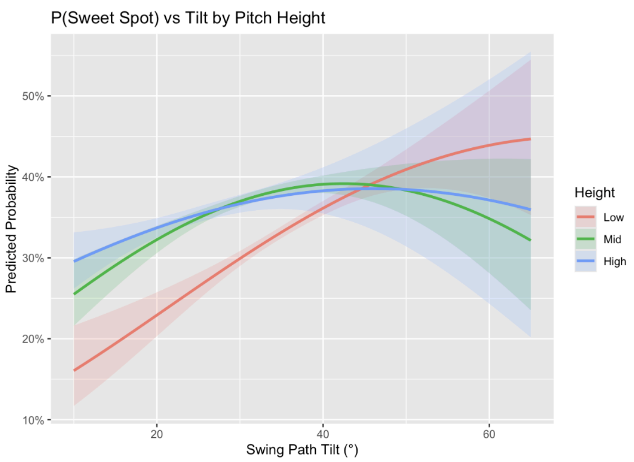

Figure 5.1. Predicted sweet spot probability as a function of swing path tilt by pitch height (low, mid, and high).

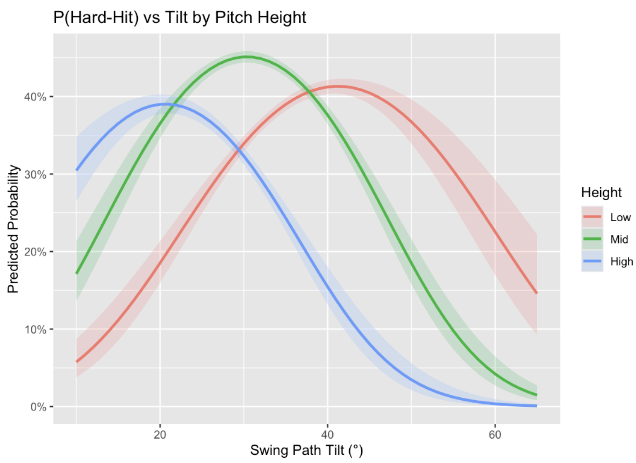

Figure 5.2. Predicted hard-hit probability as a function of swing path tilt by pitch height.

For sweet spot probability, low pitches show a largely monotonic increase as swing path tilt increases, suggesting that more upward swing paths better match the plane of low pitches. Mid-height pitches exhibit a clear inverted-U pattern, with the highest predicted sweet spot probability occurring at moderate tilt values (roughly in the 35° - 45° range). High pitches show a comparatively flatter relationship, with a mild peak around moderate tilt and a slight decline at higher tilt values, implying limited benefit from very upward swing paths against high pitches.

For hard-hit probability, the optimal tilt ranges differ more sharply across pitch heights. High pitches peak at relatively low swing path tilt (approximately 15°-25°), mid-height pitches peak around the high-20s to low-30s, and low pitches peak at higher tilt values (approximately 35°-45°). Beyond these ranges, predicted hard-hit probability declines quickly especially for high pitches indicating that overly upward swing paths are strongly disadvantageous when the pitch is located high in the zone.

Expected wOBA integrates these patterns at the outcome level. Mid-height pitches achieve the highest predicted xwOBA at moderate swing path tilt (around 35°-40°), consistent with the sweet spot results. Low pitches tend to benefit from higher tilt values, with predicted xwOBA increasing and leveling off at high tilt. In contrast, high pitches show the best xwOBA at lower tilt values and a clear decline as swing path tilt increases.

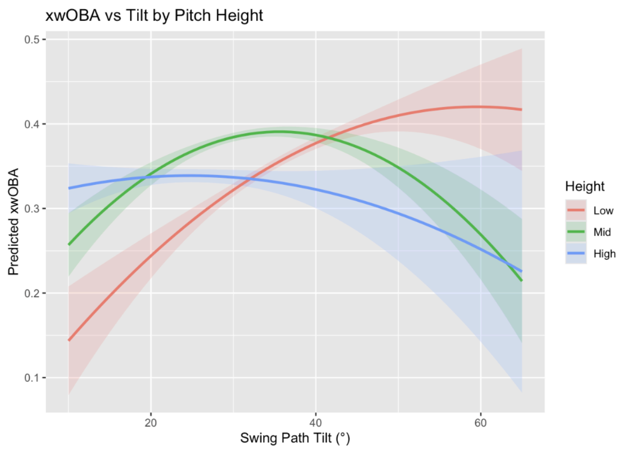

Figure 5.3. Predicted xwOBA as a function of swing path tilt by pitch height.

Taken together, the results suggest there is no single swing path tilt that is optimal across all pitch heights. Instead, the most effective swing mechanics depend on matching swing trajectory to pitch location.

6. Analysis: LASSO Regression on Runs Scored (Q4)

6.1 Model Setup

To analyze team-level offensive performance, we first started by aggregating the pitch-level data to the team-game level. Each plate appearance was assigned to the batting team, and fastballs were flagged based on pitch-type classification. For each game, we summarized runs scored along with average contact quality metrics by grouping producing one observation per team per game. Missing values were filtered, and the predictor features were standardized to ensure LASSO's regularization penalized coefficients consistently.

We defined a LASSO loss function combining mean squared error with an L1 penalty and initialized the coefficient vector at zero. Finally, we optimized this loss using the R optim function with fixed regularization parameter, yielding the extracted feature coefficients. After fitting the model, we visualized the coefficients to see which predictors were most influential and plotted residuals against predicted runs to check for consistency.

6.2 Interpretation of Results

Firstly, we can take a look at the coefficient values found by the LASSO algorithm and see that avg ISO, a value that looks at the raw power of a team largely associated with home runs, was chosen as an important stat to have for team performance alongside avg xwOBA, an all encompassing stat for hitting. All other factors according to the feature matrix

```
##     avg_launch_speed    avg_launch_angle         avg_xwOBA           avg_xBA
##        -7.517705e-05       -3.870772e-05      7.636639e-01      7.655251e-04
##              avg_ISO         barrel_rate     fly_ball_rate           BB_rate
##         8.215149e-01        5.039742e-05     -5.319460e-06      7.698551e-04
##               K_rate   avg_release_speed avg_effective_speed     fastball_rate
##        -1.110030e-03       -4.288397e-04     -4.193478e-04     -2.299941e-04
```

Figure 6.1: Table highlighting the resulting coefficients of LASSO regression.

point to other factors not being nearly as important as the previous two factors for run scoring production. As based in LASSO, these were effectively reduced to zero. This difference in importance is also reflected in the bar chart below where the two dominant features, avg xwOBA and avg ISO are highlighted from the rest of the factors.

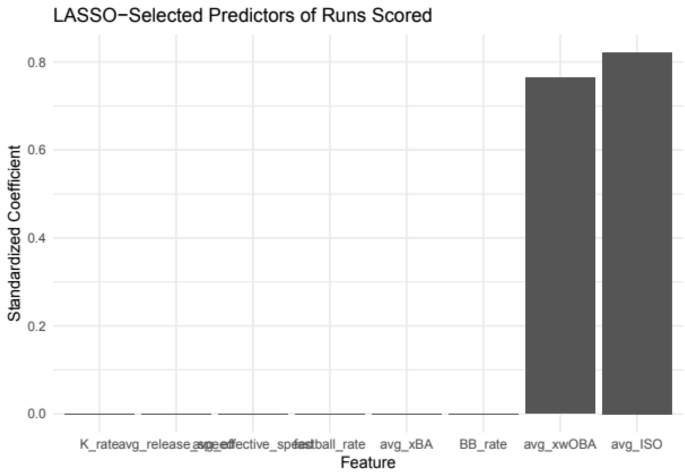

Figure 6.2: Visual demonstration of feature weights.

Using the weights from the LASSO algorithm, I plotted the actual versus predicted run values. The plot shows the general trend of team scoring, although there is some spread around the diagonal line. This means the model slightly underestimates very high-scoring games and slightly overestimates very low-scoring games. The blue line on the plot is a smoothed trend that shows if there are systematic deviations across predictors. In the residual plot, there is still some positive correlation, and some residuals are quite large. While the model may not be highly accurate in predicting runs, the goal of feature selection is achieved, as it highlights the two strongest predictors of run scoring.

Figure 6.3. Plotting LASSO predicted team runs.

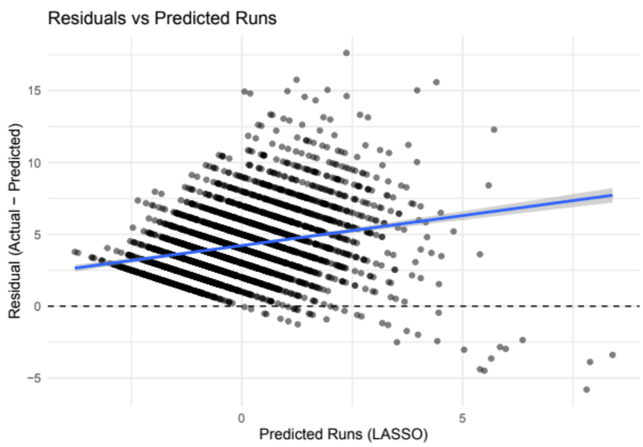

Figure 6.4. Residual plots for our predicted values.

7. Role of Course Tools

This project uses tidyverse for data preprocessing and feature engineering, ggplot2 for exploratory visualization, and regression-based models to analyze relationships between swing mechanics and contact quality. Regularization and cross-validation are applied in the team-level analysis to balance model complexity and predictive performance. Together, these tools demonstrate how course methods can be applied to real-world sports data.

8. Description of Individual Contribution

- Problem 1 code, Report: Section 3, 6.2 - Nathan Clark
- Problem 2 and 4 code, Report: Section 4, Section 6.1 - Aditya Sood
- Problem 3 code, Report: Section 0,1,2,5,7 - Yi-Chen Chang

My primary contribution to this project was the analysis of the relationship between swing mechanics and contact quality (Question 3), including data preparation, exploratory analysis, regression modeling, and visualization by pitch height.

A major challenge I encountered was beginning the modeling process before fully understanding the conceptual meaning of certain variables. Early attempts used inappropriate contact quality measures, leading to specifications that produced unrealistically strong results. Further inspection revealed issues consistent with information leakage, which required revisiting variable definitions and restructuring the analysis to ensure a proper separation between predictors and outcomes.

After correcting these issues, another challenge emerged: models based on a single swing mechanic exhibited low explanatory power. While this initially raised concerns about model validity, consistent nonlinear patterns across pitch heights and outcome measures suggested that the results were meaningful despite modest effect sizes. Through this process, I learned the importance of carefully validating variable definitions before modeling, guarding against information leakage, and interpreting statistical results in the context of noisy, observational data. I also gained experience in using visual diagnostics to assess model behavior and in focusing on stable conditional patterns rather than overall model fit when drawing substantive conclusions.

9. References

Major League Baseball. Statcast Data. Retrieved from https://baseballsavant.mlb.com/